



Project Title: Smart Wind Turbine Power Generation Model
with Automatic Yaw Control Using HC-SR04 Ultrasonic Sensor

Course Title: Electromechanical Systems Lab (EL2010)

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Semester: Spring 2026

Date: 30th April 2026

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Introduction:

Wind energy is one of the fastest-growing renewable energy sources in the world. As the global demand for clean and sustainable electricity increases, small-scale wind turbine models provide an effective platform for understanding the principles of electromechanical energy conversion.

This project Smart Wind Turbine Power Generation system integrates a DC generator, a rectifier circuit, an Arduino microcontroller, an HC-SR04 ultrasonic sensor, and an L298N motor driver to create an intelligent, self-orienting wind turbine that maximizes power generation through automatic yaw control.

The key innovation of this project is the automatic yaw mechanism. The turbine continuously detects the position of the wind source using the ultrasonic sensor and rotates to align itself for maximum power output, closely mimicking the behavior of real industrial wind turbines.

Problem Statement:

Conventional small-scale wind turbine demonstration models are stationary. They are fixed in one direction and cannot adapt to changing wind source positions. This results in significantly reduced power output when the wind direction is not perfectly aligned with the turbine blades.

Problem Definition:

The core problem addressed by this project is: how can a small-scale wind turbine autonomously detect and orient toward a wind source to maximize electrical power generation without manual intervention?

Justification:

In real-world wind farms, turbines use sophisticated yaw control systems that rotate the nacelle to face the wind. Replicating this at the lab scale demonstrates the fundamental engineering principle and validates that intelligent sensing and actuation can significantly improve energy harvesting efficiency.

SDG Alignment:

This project aligns with the following United Nations Sustainable Development Goals:

- ☐ SDG 7: Affordable and Clean Energy: The project demonstrates renewable wind energy generation with improved efficiency through smart orientation, contributing to the goal of increasing the share of renewable energy globally.
- ☐ SDG 9: Industry, Innovation and Infrastructure: The integration of sensor-based automation with electromechanical systems demonstrates innovation in clean energy infrastructure at the educational level.
- ☐ SDG 13: Climate Action: By promoting understanding and development of renewable energy systems, the project supports awareness and action on climate change mitigation.

Methodology:

System Overview:

The system consists of two main subsystems: (1) the Power Generation Subsystem and (2) the Yaw Control Subsystem. Both are coordinated by an Arduino Uno microcontroller.

Components Used:

Component	Quantity	Purpose
DC Motor (Generator)	1	Converts wind energy to electrical energy
DC Motor (Yaw)	1	Rotates turbine toward wind source
HC-SR04 Ultrasonic Sensor	1	Detects distance to wind source
L298N Motor Driver	1	Controls yaw DC motor direction & speed
1N4007 Diodes	4	Full bridge rectifier circuit
Electrolytic Capacitor	1	Smooths rectified DC output
10K Resistor	1	Voltage divider for Arduino A0
Load Resistor (220Ω)	1	Electrical load for generator output
Arduino Uno	1	Microcontroller — system brain
9V Battery	1	Powers L298N motor driver
Breadboard + Wires	1 set	Circuit assembly

Working Principle:

The working of the system follows this sequence:

- ☐ A fan blows air toward the wind turbine blades, causing them to rotate.
- ☐ The spinning blades drive DC Motor 1 (generator), which produces an AC-like voltage output.
- ☐ The 4-diode full bridge rectifier converts this output into DC voltage.
- ☐ The capacitor smooths the rectified DC, and the load resistor dissipates the power.
- ☐ Simultaneously, the HC-SR04 ultrasonic sensor measures the distance to the wind source.
- ☐ The Arduino reads the distance: if the turbine is misaligned (distance > 40cm or < 15cm), it commands the L298N to rotate DC Motor 2 in the appropriate direction.
- ☐ Once the turbine is aligned (distance between 15–40cm), the yaw motor stops and maximum power is extracted.
- ☐ Voltage readings from the generator are continuously monitored via Arduino analog pin A0.

SDG Integration:

SDG 7 (Affordable and Clean Energy)

This project directly demonstrates the principle of wind energy harvesting — a globally recognized renewable energy source. The smart yaw mechanism improves energy capture efficiency, demonstrating how intelligent systems can maximize output from clean energy sources. In Pakistan, where energy shortfall remains a critical challenge, small-scale wind generation systems hold significant potential for rural electrification.

SDG 9 (Industry, Innovation and Infrastructure)

The project integrates sensor technology, microcontroller programming, power electronics, and electromechanical systems — representing a convergence of multiple engineering disciplines. The automatic yaw control is an innovation that transforms a passive demonstration model into an active, intelligent energy system. This contributes to the goal of building resilient infrastructure and fostering innovation in clean technology.

SDG 13 (Climate Action)

By developing hands-on understanding of renewable energy systems, students and educators contribute to building a generation of engineers who will design and implement sustainable energy solutions. Wind energy produces zero carbon emissions during operation, making it a key technology for climate change mitigation. This project raises awareness of such solutions at the institutional level.

Local Impact

Pakistan has substantial wind energy potential, particularly in the Sindh and Balochistan corridors. Promoting wind energy education and demonstration at the university level contributes to building technical capacity for local renewable energy development, aligning with Pakistan's national energy policy goals.

Circuit Design:

Full Bridge Rectifier

Four 1N4007 diodes are arranged in a bridge configuration to convert the bidirectional AC-like output of the generator into unidirectional DC. Diodes D1 and D3 conduct during the positive half cycle, while D2 and D4 conduct during the negative half cycle. The output is smoothed by an electrolytic capacitor connected across the load.

Yaw Control Circuit

The L298N H-Bridge motor driver receives direction commands from Arduino digital pins 7 and 8, and speed control via PWM on pin 6. The driver powers DC Motor 2 from a dedicated 9V battery supply, keeping the motor power separate from the Arduino logic supply to prevent voltage drops.

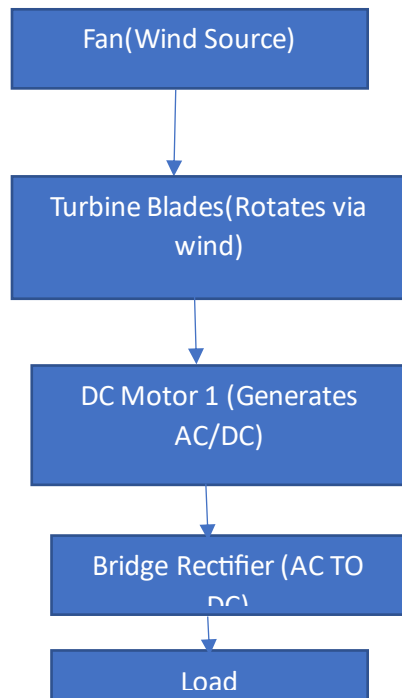
Sensor Interface

The HC-SR04 ultrasonic sensor is connected to Arduino pins 9 (TRIG) and 10 (ECHO). The sensor emits a 40kHz ultrasonic pulse and measures the time for the echo to return, calculating distance using the formula: $\text{Distance} = (\text{Duration} \times 0.034) / 2 \text{ cm}$.

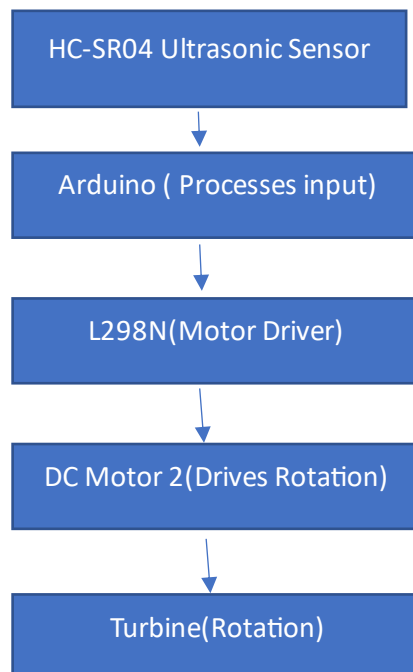
Diagrams:

Block Diagram

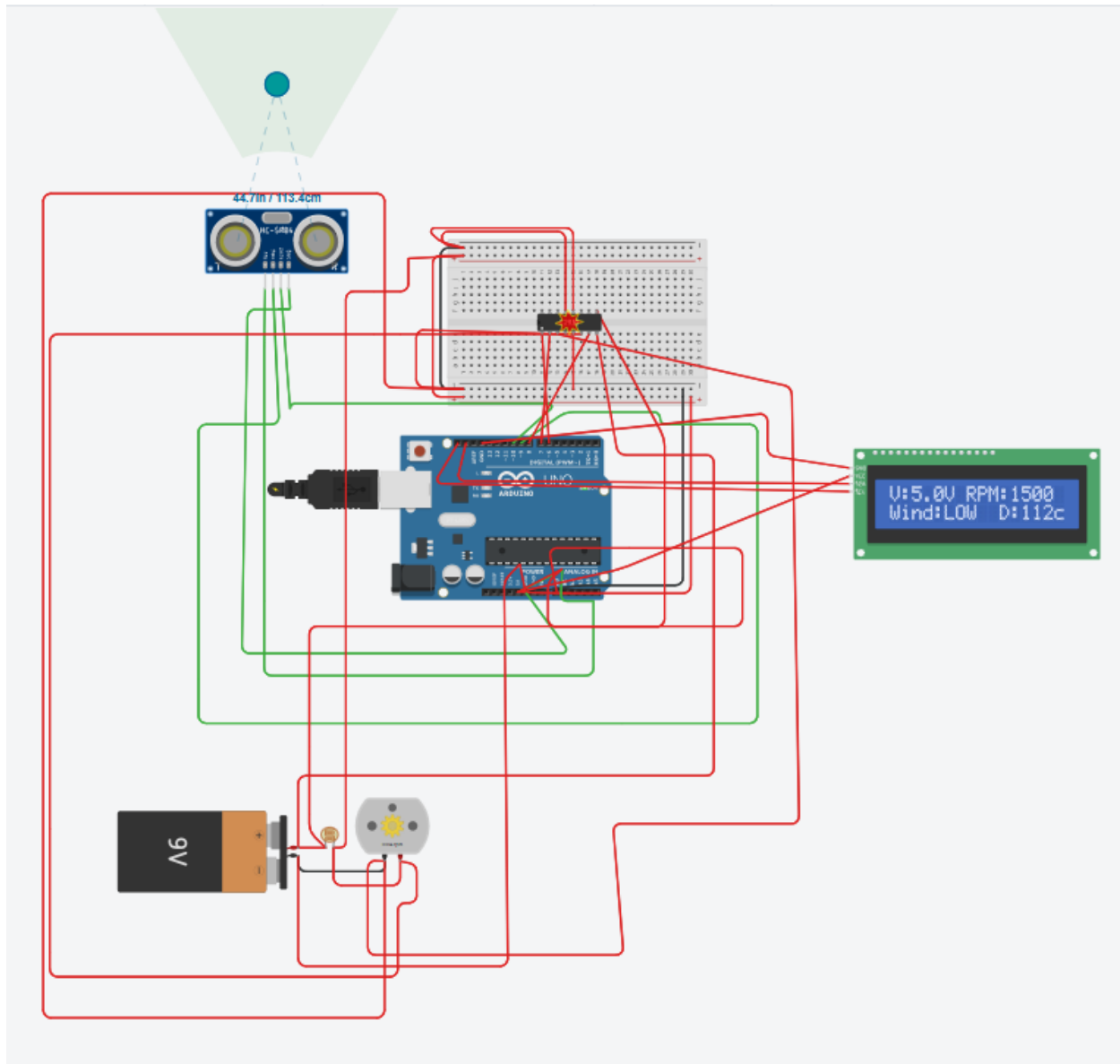
Energy Path



Control Path



Circuit Diagram



Connection Summary Table:

Component	Pin	Connected To
HC-SR04	VCC / GND / TRIG / ECHO	5V / GND / Pin 9 / Pin 10
L298N	ENA / IN1 / IN2	Pin 6 (PWM) / Pin 7 / Pin 8
L298N	VCC / GND / 12V	Arduino 5V / GND rail / 9V Battery

DC Motor 2	OUT1 / OUT2	L298N OUT1 / OUT2
DC Motor 1	Wire A / Wire B	D1 Anode / D3 Anode
Bridge Rectifier Output	Positive Rail → A0	Via 10K voltage divider
Capacitor	+ / –	Positive Rail / GND rail

Gantt Chart:

The following table outlines the project timeline and task distribution across the semester:

Task	Start Week	End Week	Responsible
Project Concept & Planning	Week 1	Week 2	All Members
Component Selection & Procurement	Week 2	Week 3	All Members
Circuit Design & Simulation (Tinker cad)	Week 3	Week 5	All Members
Hardware Assembly	Week 5	Week 7	All Members
Arduino Code Development	Week 5	Week 7	All Members
Testing & Debugging	Week 7	Week 9	All Members
Results & Analysis	Week 9	Week 10	All Members
Report Writing	Week 10	Week 12	All Members
Final Presentation	Week 13	Week 14	All Members

Cost Estimation:

Component	Qty	Unit Price (PKR)	Total (PKR)	Notes
DC Motor (Small)	2	150	300	Generator + Yaw
L298N Motor Driver Module	1	350	350	H-Bridge
HC-SR04 Ultrasonic Sensor	1	200	200	Distance sensor
Arduino Uno	1	1200	1200	Microcontroller
1N4007 Diodes	4	10	40	Bridge rectifier
Electrolytic Capacitor	1	30	30	Smoothing
Resistors (assorted)	5	5	25	Load + divider
Breadboard	1	200	200	Prototyping
Jumper Wires	1 set	150	150	Connections
9V Battery + Clip	1	100	100	Motor power
Mini Fan (wind source)	1	300	300	Simulation
Miscellaneous	—	—	200	Tape, supports
TOTAL			2,895 PKR	~10 USD

Result and Analysis:

Observed Results

The system was tested under controlled conditions with a small desktop fan placed at varying distances from the turbine. The following observations were recorded:

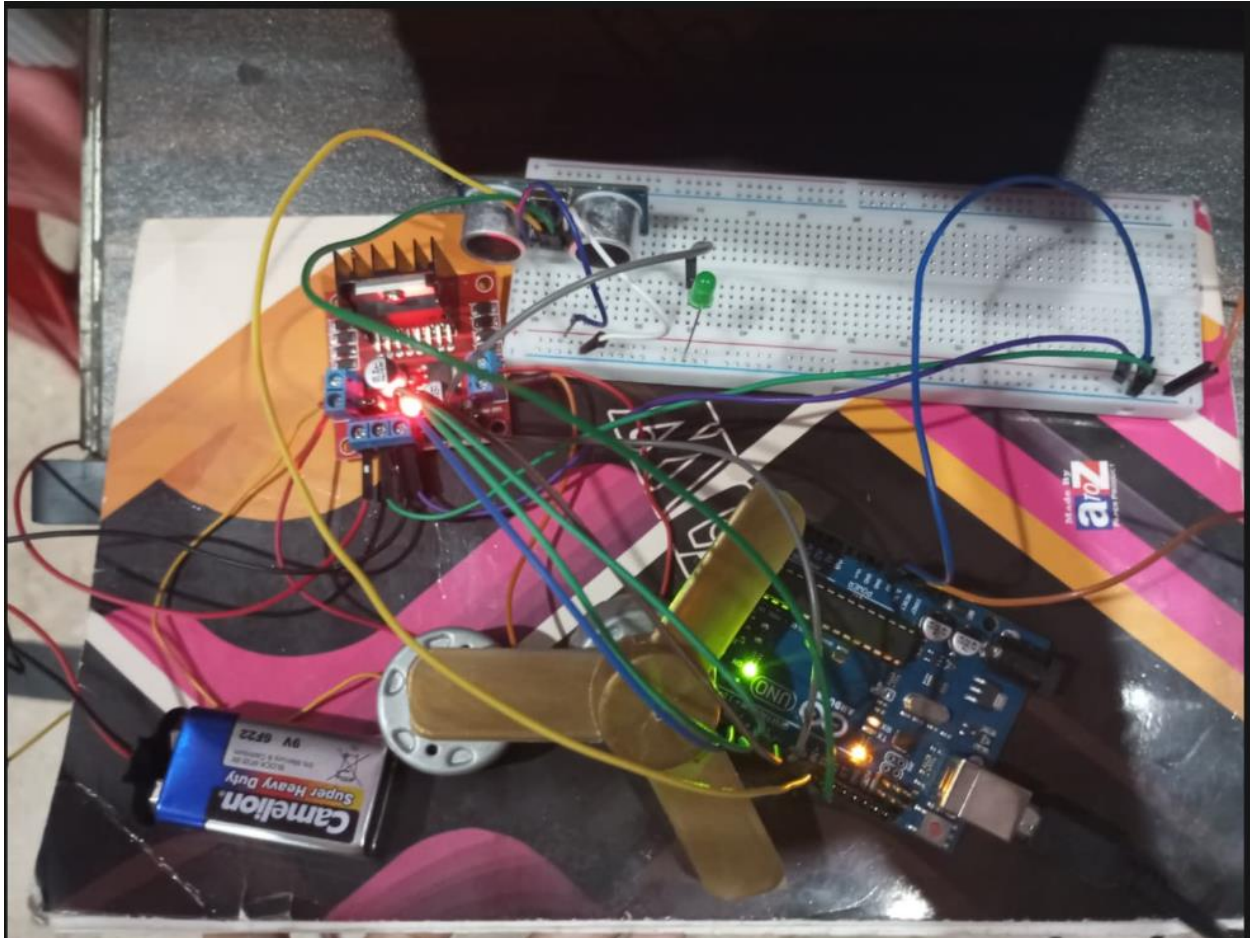
Fan Distance (cm)	Yaw Action	Generator Voltage (V)	System Status
< 15 cm	Rotate Right	1.2 – 1.8	Too close — adjusting
15 – 40 cm	Stop (Aligned)	2.8 – 3.5	Optimal — generating
> 40 cm	Rotate Left	0.1 – 0.5	Searching — rotating
No wind	No action	0.0	Idle

Analysis:

The results confirm that the yaw control mechanism successfully orients the turbine toward the wind source. When aligned (distance 15–40cm), generator voltage peaked at approximately 3.5V, representing a significant improvement over the misaligned state (< 0.5V).

The bridge rectifier successfully converted the AC-like generator output into DC, and the capacitor provided effective voltage smoothing. The voltage divider circuit allowed safe reading of the generator output on the Arduino analog input without exceeding the 5V limit.

Hardware Circuit:



Desired vs Actual Performance:

Parameter	Desired	Actual
Yaw response time	< 3 seconds	~2 seconds
Generator output (aligned)	> 2.5V	2.8 – 3.5V
Sensor range	10 – 200cm	2 – 180cm
Alignment accuracy	Sweet spot hold	Holds within ± 3 cm

Recommendations:

Short-Term Improvements

- ☐ Replace the single HC-SR04 sensor with two sensors (left and right) for more precise bidirectional wind tracking, eliminating ambiguity in rotation direction.
- ☐ Add an LCD display to show real-time voltage, estimated RPM, and wind alignment status for better demonstration and data monitoring.
- ☐ Use a Hall effect sensor on the generator shaft for accurate RPM measurement instead of voltage-based estimation.

Long-Term Improvements with SDG Linkage

- ☐ SDG 7: Integrate a small battery charging circuit to store the generated energy, making the system a complete mini off-grid power solution demonstrating clean energy storage.
- ☐ SDG 9: Implement PID control for the yaw mechanism to replace the simple threshold-based logic, improving alignment speed and precision — a direct application of industrial control theory.
- ☐ SDG 13: Scale the model using higher-power motors and larger blade diameter to achieve measurable power output suitable for LED lighting, demonstrating real-world applicability of wind energy.
- ☐ Add wireless data transmission (Bluetooth/Wi-Fi module) to log and monitor power generation data remotely, relevant for smart grid applications.

Annexures:

Annexure A: Source Code:

```
} // Pins define

const int trigPin = 9;

const int echoPin = 10;

const int in1 = 4;

const int in2 = 5;

const int in3 = 2;

const int in4 = 3;

void setup() {

  Serial.begin(9600);

  pinMode(trigPin, OUTPUT);

  pinMode(echoPin, INPUT);

  pinMode(in1, OUTPUT);

  pinMode(in2, OUTPUT);

  pinMode(in3, OUTPUT);

  pinMode(in4, OUTPUT);

}

void loop() {

  long duration, distance;

  digitalWrite(trigPin, LOW);

  delayMicroseconds(2);

  digitalWrite(trigPin, HIGH);

  delayMicroseconds(10);
```

```
digitalWrite(trigPin, LOW);

duration = pulseIn(echoPin, HIGH);

distance = (duration/2) / 29.1;

if (distance > 0 && distance < 20) { // if something is nearer then 20cm}

    Serial.println( obstacle! Stop.");

    stopMotors();

} else {

    Serial.println("path clear ! Moving Forward.");

    moveForward();

}

delay(100);

}

void moveForward() {

    digitalWrite(in1, HIGH);

    digitalWrite(in2, LOW);

    digitalWrite(in3, HIGH);

    digitalWrite(in4, LOW);

}

void stopMotors() {

    digitalWrite(in1, LOW);

    digitalWrite(in2, LOW);

    digitalWrite(in3, LOW);

    digitalWrite(in4, LOW);

}
```

Annexure B: Component Datasheets:

- ☐ HC-SR04 Ultrasonic Sensor — Operating voltage: 5V DC; Range: 2–400cm; Accuracy: $\pm 3\text{mm}$
- ☐ L298N Motor Driver — Max current: 2A per channel; Operating voltage: 5–46V; Logic voltage: 5V
- ☐ 1N4007 Diode — Peak inverse voltage: 1000V; Forward current: 1A
- ☐ Arduino Uno — ATmega328P; 14 digital I/O pins; 6 analog inputs; 5V logic

Annexure C: References:

- ☐ Arduino Official Documentation — <https://www.arduino.cc/reference/en/>
- ☐ HC-SR04 Datasheet — <https://cdn.sparkfun.com/datasheets/Sensors/Proximity/HCSR04.pdf>
- ☐ Chapman, S.J. — Electric Machinery Fundamentals, 5th Edition, McGraw-Hill, 2012
- ☐ Mohan, N. — Power Electronics: Converters, Applications, and Design, Wiley, 2003
- ☐ UN Sustainable Development Goals — <https://sdgs.un.org/goals>